

Data Structures and Algorithms

1. Data Structures and Algorithms

Data Structures are ways of storing and organizing data. Algorithms are step-by-step instructions used to solve problems or work with data. DSA helps programs handle large amounts of data more efficiently.

2. Linear Data Structure

A linear data structure stores data in a sequential order, where elements are arranged one after another. Examples include arrays, linked lists, stacks, and queues.

3. One-Dimension Array

A one-dimensional array is a data structure used to store multiple elements in a single sequence. Each element can be accessed using its position or index.

4. The Big O Notation

Big O notation is used to describe how the running time of an algorithm increases as the amount of data increases. Examples include $O(1)$, $O(n)$, $O(n^2)$, and $O(\log n)$.

5. Basic Searching Algorithms

Searching algorithms are used to find a specific value in a collection of data. Two basic searching algorithms are Linear Search and Binary Search.

6. Linear Search

Linear Search checks the elements one by one from the beginning until the target value is found. Its time complexity is $O(n)$ in the worst case. It does not require the array to be sorted.

7. Binary Search

Binary Search searches for a value by repeatedly dividing a sorted array into two parts. It is faster than Linear Search and has a time complexity of $O(\log n)$.